Individual Visualisation Report, Student Number: 25135657, Word Count: 493

Mapping London's Path to Vision Zero: A Multi-Scalar Traffic Collisions Analysis (2017-2025)

Overview

Transport for London is currently advancing its **"Vision Zero"** action plan, aiming to eliminate all deaths and serious injuries from London's transport network by 2041. In alignment with this goal, this project provides a comprehensive interactive platform for exploring road traffic collisions across Greater London from 2017 to 2025.

The visualisation employs a multi-scalar approach to deliver a "macro-to-micro" perspective. At the strategic level, the map presents **Collision Risk Density** across two administrative scales: Local Authority (LAD) and Middle Layer Super Output Area (MSOA). As the user zooms into high-risk areas, the map transitions into a granular display of **individual collision points**, classified by severity. This allows planners and residents to first identify broader safety trends before diving into specific high-risk locations, balancing strategic policy oversight with community-level insights(Cockburn, 2009).

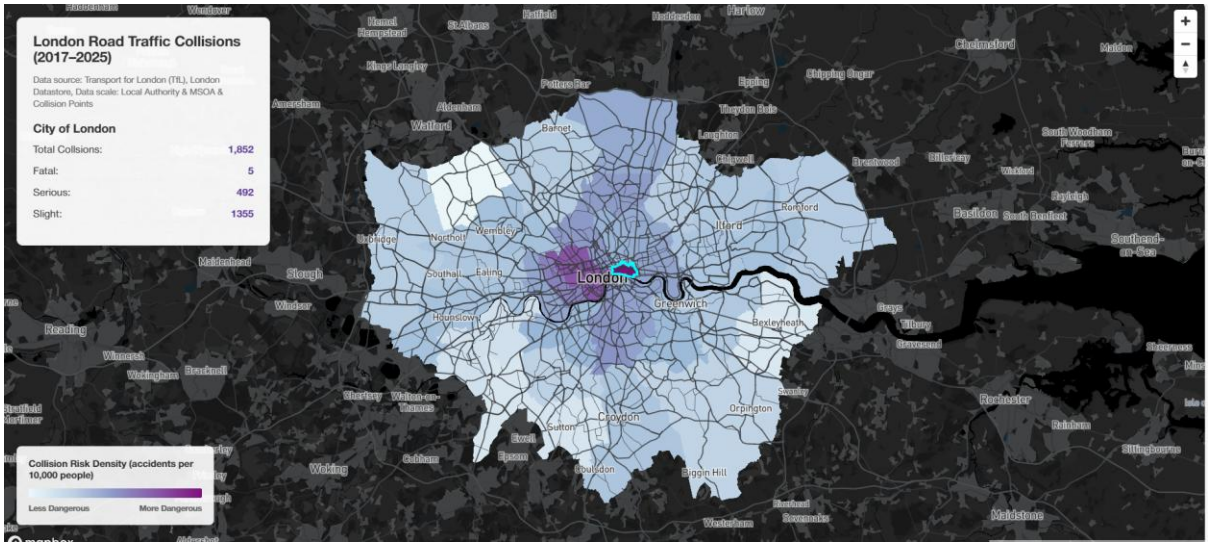

Figure 1: The London traffic collisions Interactive Map at Local Authority Scale

Technical Aspects: Data and Methods

The project includes three data sources:

- **Road Safety Data** from Transport for London (TfL). [Road safety data - Transport for London]
- **Administrative Boundary** from London Datastore. [Statistical GIS Boundary Files for London - London Datastore]
- **Population Data** from London Datastore. [Super Output Area Population (LSOA, MSOA), London - London Datastore]

A timeframe from **2017 to 2025** was selected to capture long-term safety trends, smoothing out annual anomalies (such as the atypical traffic patterns during the COVID-19 pandemic).

Data preparation in QGIS involved joining over **237231 incident records** with population-weighted administrative polygons. A critical technical step was to compute **"Collision Risk Density"** (accidents per 10,000 people). This reduces the bias in results caused by population differences across regions (Fridman *et al.*, 2019).

The **Mapbox GL JS** engine manages the seamless transition between layers:

- **LAD Layer (Zoom 0–10):** Provides a borough-wide strategic overview.
- **MSOA Layer (Zoom 10–22):** Reveals local community variances.
- **Point Data Layer (Zoom 13+):** Identifies the exact coordinates and severity of each crash.

Interactive functionality allows the dashboard to update in real-time. When hovering over different administrative zones, the panel displays the **Total Collision Count** and the specific breakdown of **Fatal, Serious, and Slight** collisions. When displaying accident points, it is combined with a drop-down filtering function to achieve precise filtering of accident severity levels

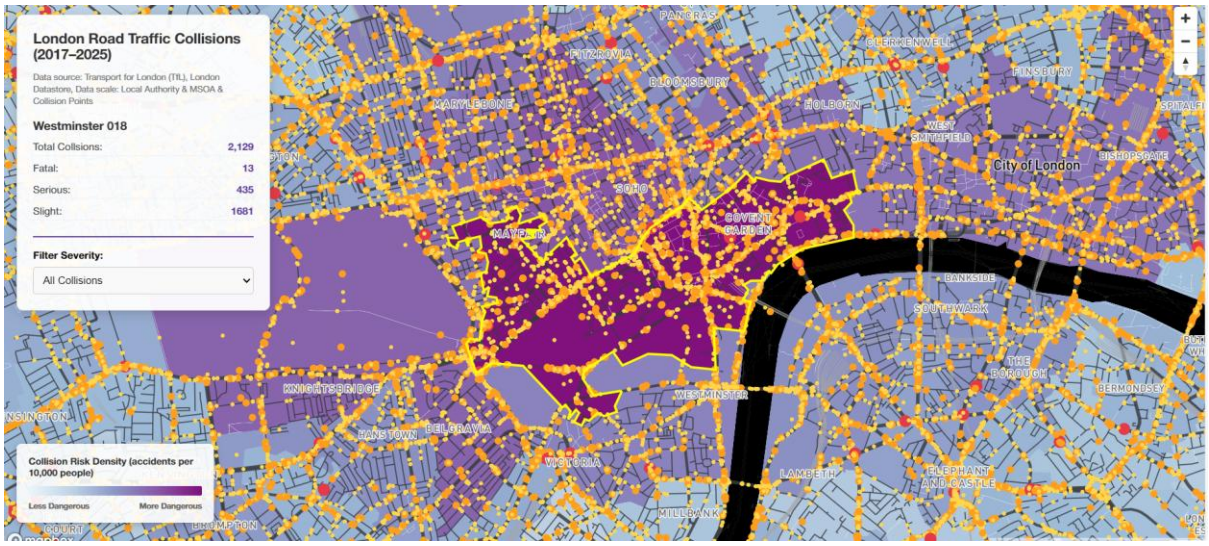

Figure2. Distribution of All Collision Points at a High Zoom Level

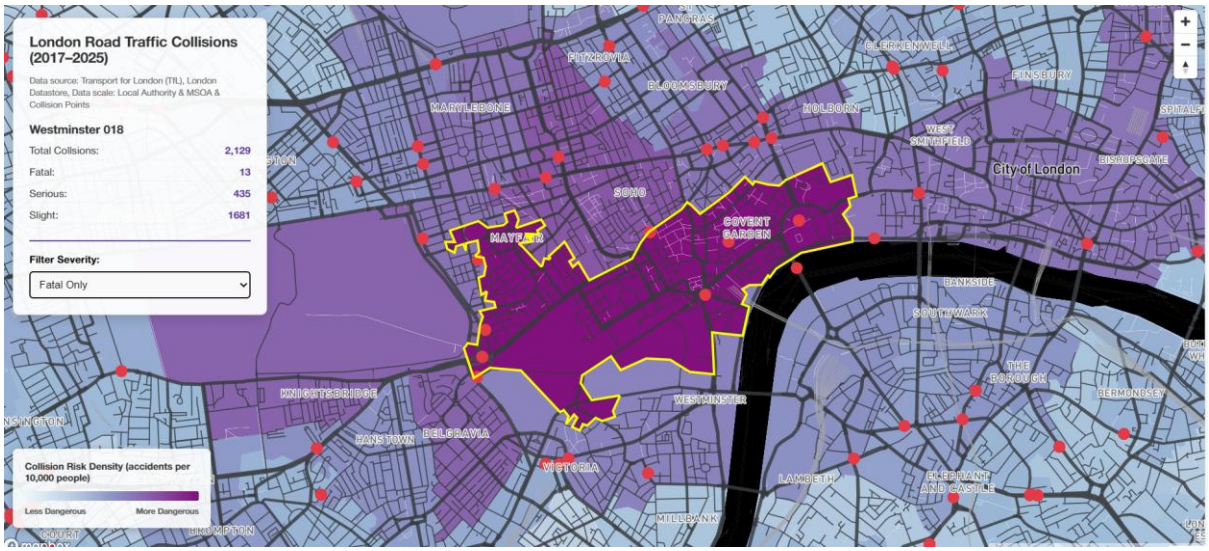

Figure3. Distribution of Fatal Collision Points at a High Zoom Level

Cartographic Aspects

The map utilizes a sequential color scheme referencing ColorBrewer (ColorBrewer: Color Advice for Maps), with deep purple indicating the highest collision densities. Point data is rendered with warm-toned highlights. To visually prioritize the most critical incidents, the diameter of Fatal collision circles is set to the maximum scale.

Limitations and Reflection

Data accuracy is limited by the rounding of coordinates to the nearest junction, which may obscure specific district risks. Future research should integrate a time slider and infrastructure data to better analyze the causal factors behind identified hotspots.

References

Cockburn, A. (2009) 'A review of overview+detail, zooming, and focus+context interfaces | ACM computing surveys', *ACM Computing Surveys (CSUR)* [Preprint]. Available at: https://dl.acm.org/doi/10.1145/1456650.1456652 (Accessed: 8 February 2026).

Fridman, L. *et al.* (2019) 'Driver and road characteristics associated with child pedestrian injuries', *Accident Analysis & Prevention*, 131, pp. 248–253. Available at: https://doi.org/10.1016/j.aap.2019.07.007.